## QUESTIONS 3

1. Solve the following system by applying the Gaussian Elimination method

$$\begin{aligned} 2x_1 - x_2 + 2x_3 &= 8 \\ 2x_1 - x_2 - x_3 + x_4 &= 5 \\ 2x_1 + 2x_2 + x_3 &= 4 \\ 2x_1 - x_2 + 2x_3 + 2x_4 &= 10 \end{aligned}$$

2. Solve the following system by applying the Cramer's Rule

$$\begin{aligned} 2x_1 + 3x_2 - x_3 &= 4 \\ x_1 - 2x_2 + 2x_3 &= 6 \\ -x_1 + 4x_2 + x_3 &= 5 \end{aligned}$$

3. Solve the following system by applying the LU-Factorization method

$$\begin{aligned} 2x_1 - x_2 + x_3 &= -1 \\ 3x_1 + 3x_2 + 9x_3 &= 0 \\ 3x_1 + 3x_2 + 5x_3 &= 4 \end{aligned}$$

4. Let $A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 6 & 2 \\ 2 & 2 & 5 \end{bmatrix}$.

a) Show that $A$ is a positive definite matrix.

b) Find a Cholesky factorization of $A$.

c) Use this Cholesky factorization to solve the system $Ax = \begin{bmatrix} 4 & 6 & -1 \end{bmatrix}^T$

5. Evaluate the eigenvalues, eigenvectors and spectral radius of the following matrices:

$$A_1 = \begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 7 \end{bmatrix}$$